Population exposure and background weighting in reported disease occurrence

Results

Population exposure and background sampling can generate urban-cover and healthcare-access associations even when individual infection risk and reporting are constant. In reconstructed dengue data, changing logarithmic to linear population weights, with cell area and the inhabited-cell frame fixed, reduced urban coefficients and reversed travel-time posterior means in all four paired draws. These directions persisted after omitting five unresolved predictors and under separate country exclusions. Sixteen additional pairs showed the same directions for the complete-recipe contrast. Urban associations remained positive. The results show why Gibb et al.'s detection-related coefficients cannot uniquely identify reporting effects or establish their causal importance relative to environmental processes.

Table 1. Median paired changes in posterior means, with observed minima and maxima in brackets. The controlled log-to-linear comparison uses four pairs; the complete-recipe comparison uses 16 additional pairs. Coefficients are log-odds per archived predictor standard deviation; travel time is log(1 + mean travel minutes) before scaling. Ranges describe background-draw variation, not credible or confidence intervals.

Weighting contrast	Urban change	Travel-time change
Controlled log to linear (4 pairs)	−1.274 [−1.435, −0.999]	0.756 [0.668, 0.797]
Complete recipe (16 new pairs)	−1.250 [−1.499, −0.772]	0.816 [0.703, 0.979]

Population exposure and occurrence associations

With constant individual infection risk and reporting, logarithmic population weighting produced positive urban and negative healthcare-travel coefficients (Figure 1). Population-proportional weighting brought both coefficients close to zero when occurrence was rare. Under saturation, population weights no longer matched the expected distribution of occurrence; the known-probability comparator remained centered near zero.

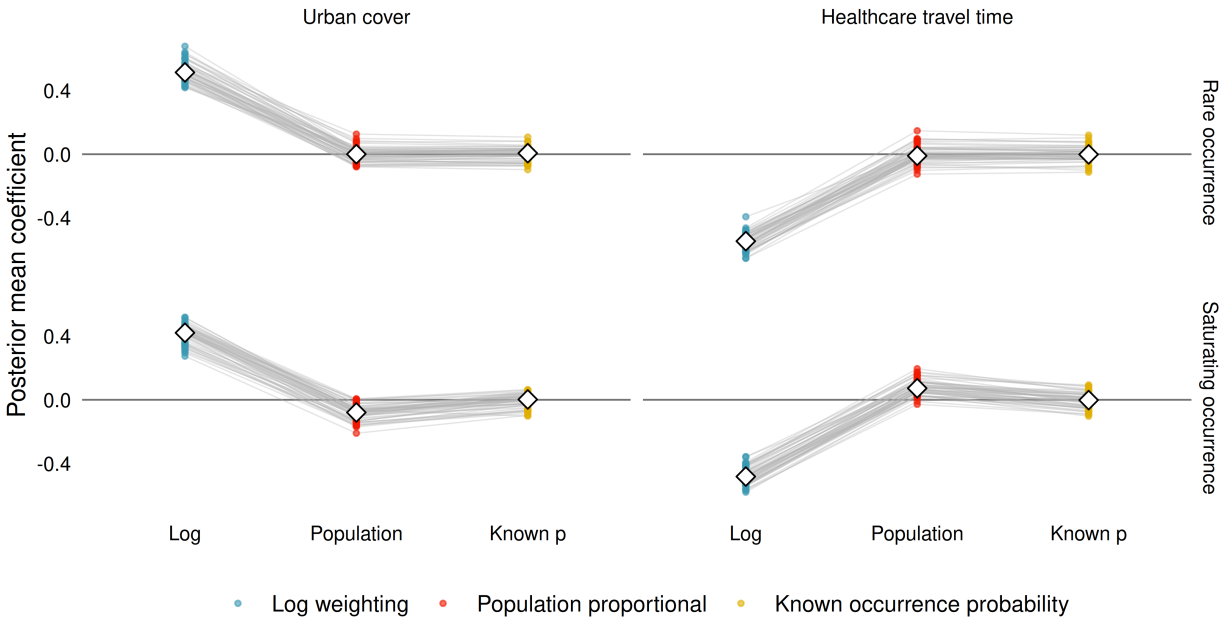


Figure 1. Simulations with constant individual risk and reporting. Points are replicate posterior means; gray lines connect paired fits, and black diamonds mark medians. The known-p comparator samples backgrounds in proportion to the generating occurrence probability. Each scenario has 50 draws on the same landscape.

The probability of at least one reported case increases with population even when individual risk and reporting are constant. Population-proportional backgrounds approximately account for this relationship when occurrence is rare. Logarithmic weights leave a residual population gradient, while saturation makes even linear population weights an imperfect comparator.

Dengue coefficients under alternative backgrounds

The dengue comparison used the same 11,313 occurrence records and reconstructed covariates in every fit, with 12,108 backgrounds per arm. The implemented recipe weights cells by area × log(1 + population), with a positive weight for empty cells. The alternative weights cells by population count. Covariate scaling, mesh and priors were held fixed.

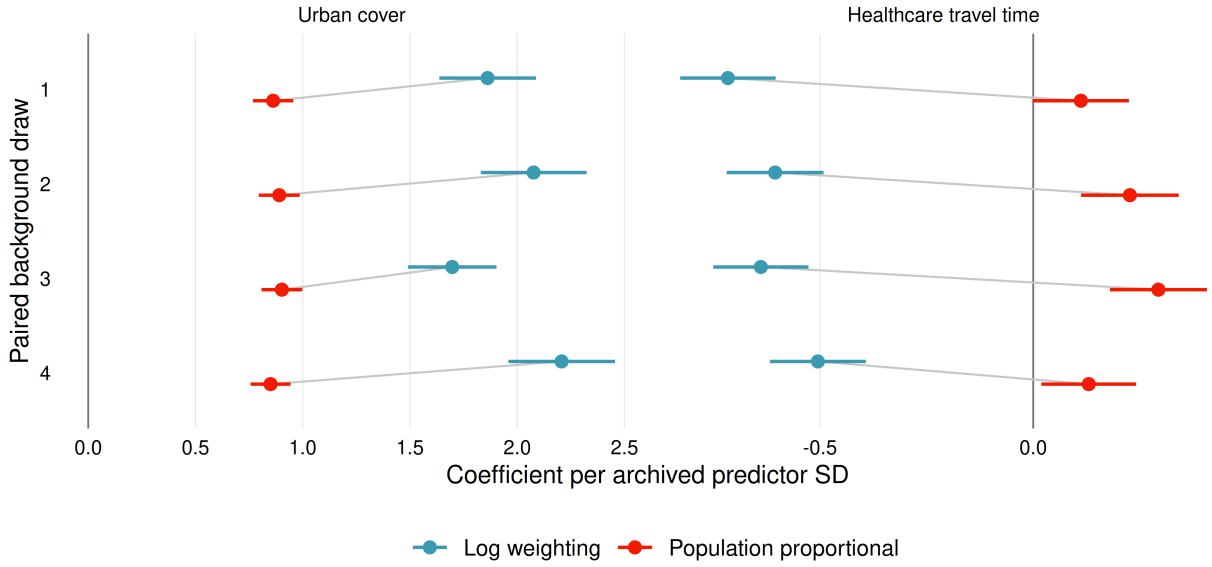


Figure 2. The four original endpoint comparisons for dengue. Points and horizontal intervals show posterior means and individual-model 95% credible intervals. Gray lines connect paired fits. Blue denotes the implemented log recipe; red denotes population-proportional weighting. Draws 1–4 correspond to seeds 980019, 980027, 980035 and 980043. Panel scales differ; intervals are not intervals for the paired difference.

All eight urban credible intervals remained positive. Healthcare-travel means reversed in all four pairs; under population weighting, three intervals were positive and one included zero. Paired urban changes ranged from −1.357 to −0.794, and travel changes from +0.635 to +0.932.

Separating the sampling choices

Four sampling arms separate the recipe into conditional changes. A uses area × log(1 + population), assigning empty cells the median positive log weight. B retains the area factor and logarithm but excludes empty cells. C changes B to linear population weighting. D removes the area factor, leaving population-proportional weights.

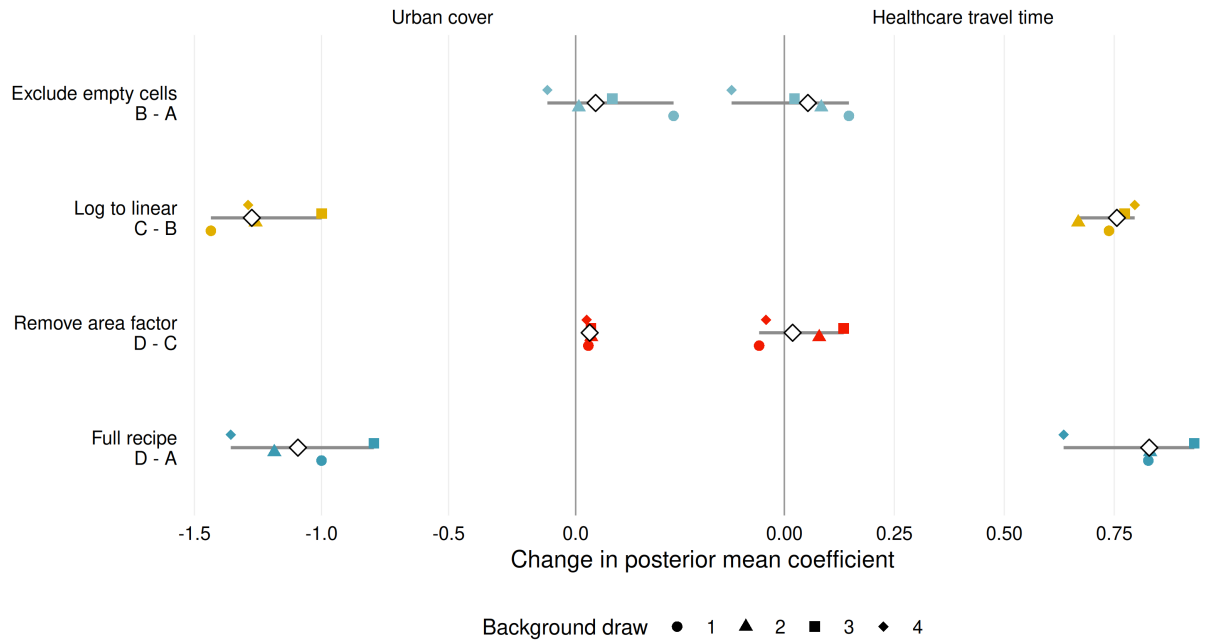


Figure 3. Changes along the sampling sequence. Colored points show the four paired differences in posterior means, white diamonds their medians, and gray segments their observed ranges. C minus B changes the logarithm with area and the inhabited-cell frame fixed; D minus A changes the complete recipe. The intermediate changes depend on their position in this sequence and are not factorial main effects. Their values sum to the total within each draw; their medians need not sum.

The log-to-linear step produces the largest shift in this sequence: urban changes range from −1.435 to −0.999, and travel-time changes from 0.668 to 0.797. Changes from the empty-cell rule are smaller and vary in direction. Removing area weighting after linear weighting also produces smaller changes.

Reconstruction checks at archived locations

Replacing archived covariates with reconstructed values at the same case and background locations changed the focal coefficients very little (Table 2). Restoring archived values for the five unresolved predictors, individually and together, also produced little change in urban and travel-time estimates (Figure 4).

Table 2. Posterior means at the same archived locations. The full-precision comparison reports all 13 coefficients, including the intercept. Reconstruction has larger effects on some other coefficients, including temperature change (−0.089), forest loss (−0.032) and protected areas (−0.029).

Predictor	Archived	Reconstructed	Change
Urban cover	1.918	1.919	0.001
Healthcare travel time	−0.686	−0.674	0.012

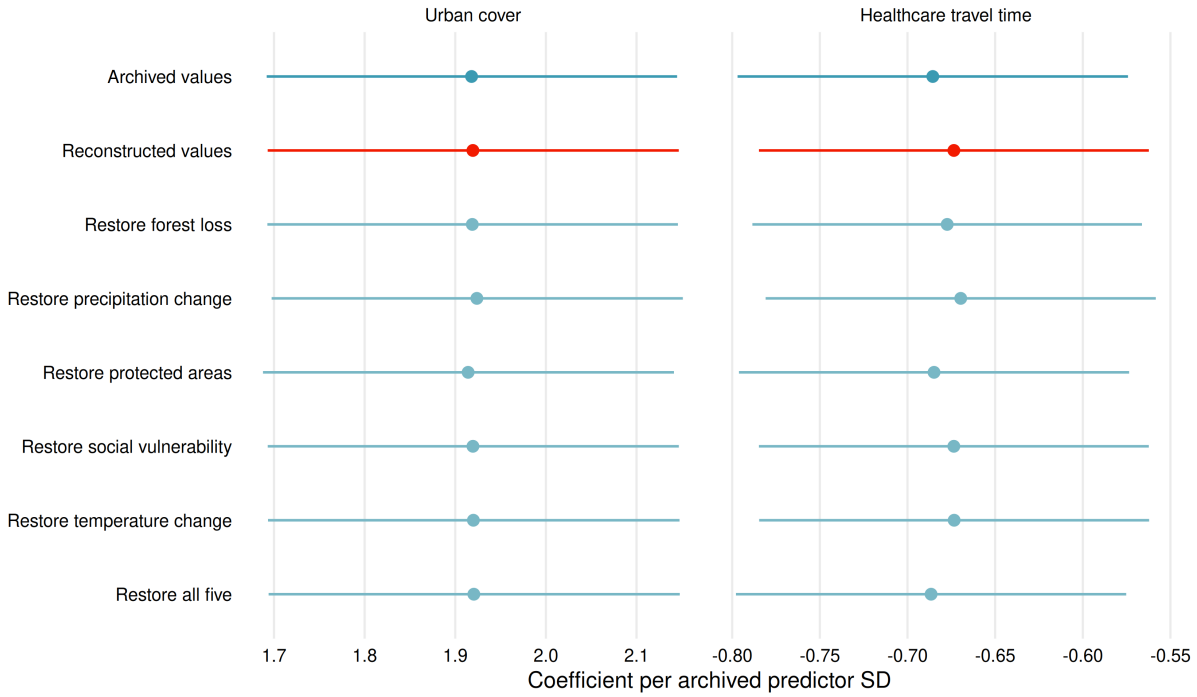


Figure 4. Archived and reconstructed fits, followed by restoration of archived values for forest loss, precipitation change, protected areas, social vulnerability and temperature change, individually and together. Each restoration applies to both cases and backgrounds at the original locations. Points and segments are posterior means and marginal 95% credible intervals. Panel scales differ.

Weighting contrasts after omitting unresolved predictors

The controlled weighting contrast persisted after omitting the five predictors with unresolved source or processing differences: forest loss, precipitation change, protected areas, social vulnerability and temperature change (Table 3; Figure 5). The reduced models retained the other seven predictors, including urban expansion, with the same observations, backgrounds and model settings.

Table 3. Median controlled log-to-linear changes and four-draw ranges. Travel-time posterior means reverse sign in all four pairs under either adjustment set; urban cover remains positive. Omitting predictors tests sensitivity to the adjustment set.

Adjustment set	Urban change	Travel-time change
All 12 predictors	−1.274 [−1.435, −0.999]	0.756 [0.668, 0.797]
Five unresolved predictors omitted	−1.426 [−1.622, −1.151]	0.796 [0.749, 0.831]

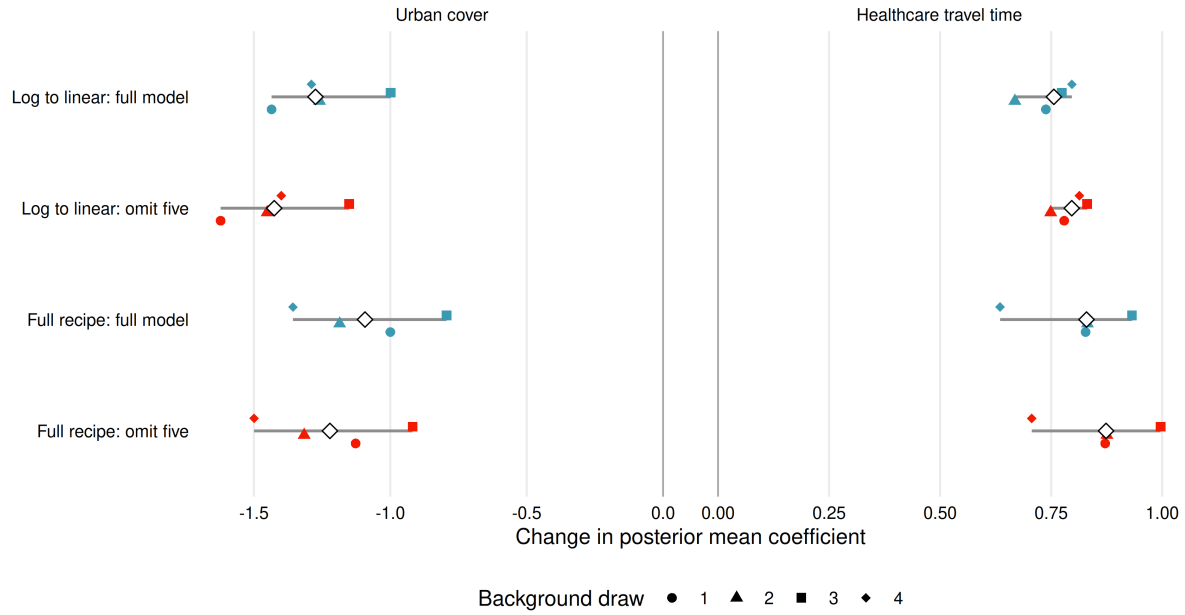


Figure 5. Controlled log-to-linear and complete-recipe contrasts with all 12 predictors and after omitting the five unresolved predictors. Colored points show paired differences in posterior means; white diamonds and gray segments show medians and observed four-draw ranges. These are not posterior intervals for the contrasts.

Geographical influence

Urban coefficients decreased and travel-time posterior means reversed from negative to positive after separately excluding Brazil, India and the Philippines. These directions held in all 12 paired comparisons for each weighting contrast (Table 4; Figure 6).

Table 4. Median paired changes and ranges across four sampled backgrounds after each coordinate-based country omission. Country labels were missing in the original records. These exclusions test the influence of modeled geography, not independently validated country of infection or reporting source. The changed geography and sample composition can affect coefficient magnitude and precision.

Controlled log to linear

Country omitted	Urban change	Travel-time change
Brazil	−1.287 [−1.490, −0.954]	0.698 [0.609, 0.734]
India	−1.307 [−1.519, −1.040]	0.815 [0.711, 0.832]
Philippines	−1.293 [−1.465, −0.982]	0.781 [0.704, 0.821]

Complete recipe

Country omitted	Urban change	Travel-time change
Brazil	−1.107 [−1.343, −0.823]	0.706 [0.475, 0.814]
India	−1.165 [−1.412, −0.856]	0.885 [0.705, 0.998]
Philippines	−1.110 [−1.362, −0.797]	0.861 [0.643, 0.967]

Under arms C and D, urban credible intervals were positive after every country exclusion. Travel-time intervals were positive in two of four fits without Brazil and all four without India or the Philippines. These comparisons reuse draws and overlapping records.

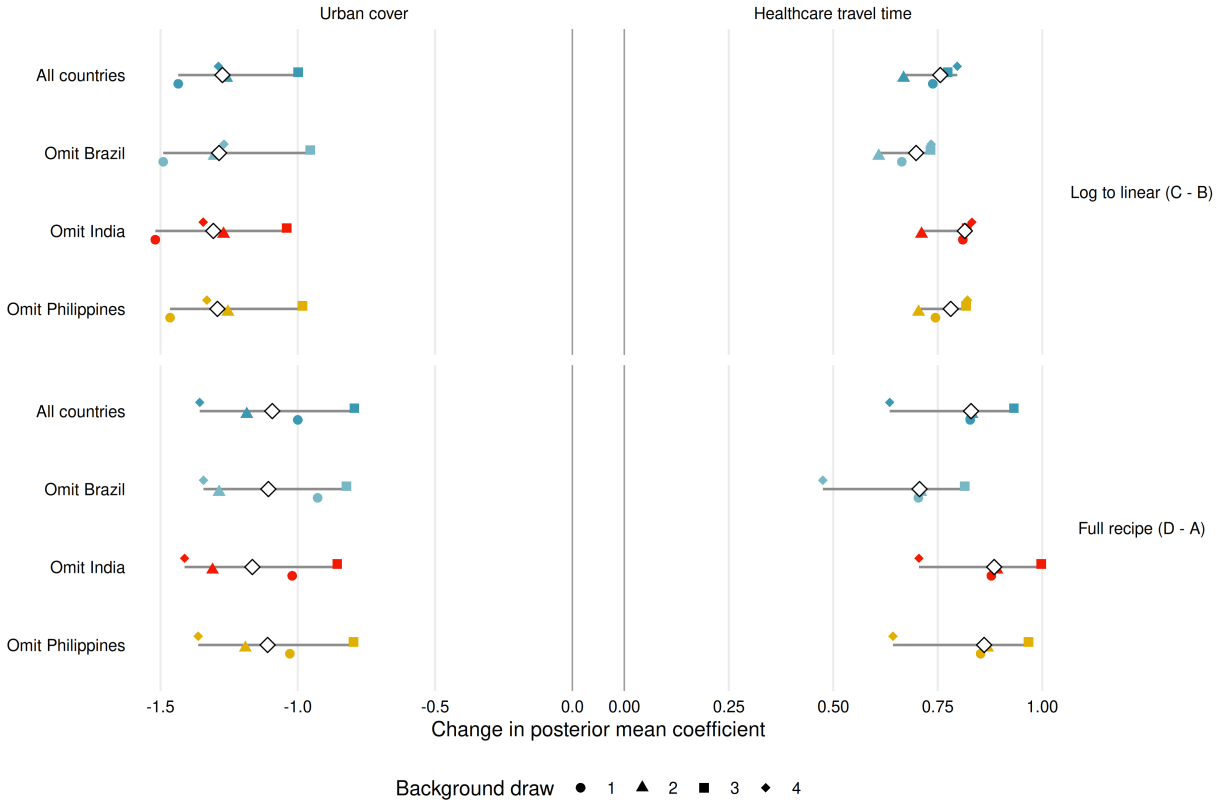


Figure 6. Changes in posterior means for the two weighting contrasts. All countries is the global reference; the other rows restrict those same draws to exclude coordinates within each named country. Draw-level changes and their observed ranges describe variation over four sampled backgrounds, with arm-dependent retained background counts after restriction.

Additional background draws

All 16 additional pairs showed lower urban coefficients and a reversal of travel-time means from negative to positive under population-proportional sampling (Table 5; Figure 7). These pairs quantify variation in the complete-recipe contrast while holding cases, reconstructed covariates and model settings fixed.

Table 5. Summary of the 16 additional complete-recipe pairs. Draw SD is the sample standard deviation of the paired changes. Mean MCSE is SD divided by the square root of 16; it measures Monte Carlo precision of the average change, conditional on the fixed cases, surfaces and model. It is not a posterior standard deviation or an uncertainty interval for a coefficient contrast.

Predictor	Mean change	Draw SD	Mean MCSE	Median	Range
Urban cover	−1.245	0.176	0.044	−1.250	[−1.499, −0.772]
Healthcare travel time	0.816	0.071	0.018	0.816	[0.703, 0.979]

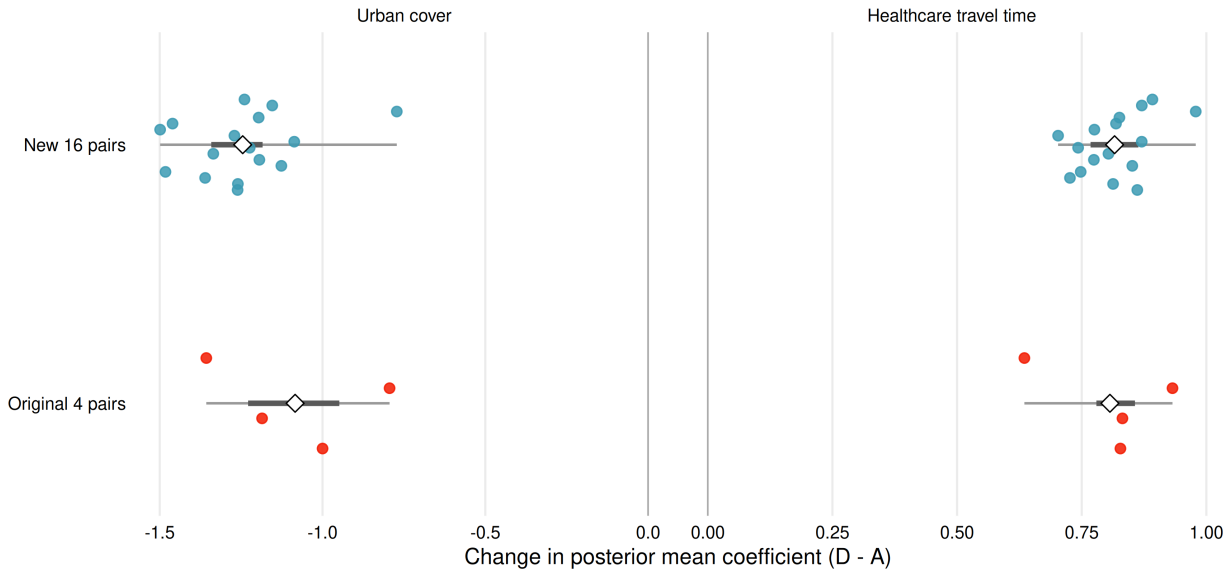


Figure 7. Complete-recipe changes in the 16 new pairs and four original reference pairs. Points are within-pair differences between posterior means, D minus A. White diamonds show mean changes, thick segments the empirical 25th–75th percentiles, and thin segments the observed range. These summarize background-draw variation for the complete recipe; the controlled B/C comparison remains based on four pairs.

Urban credible intervals were positive in all 32 additional fits. Travel-time intervals were negative in all 16 log-recipe fits; under population weighting, 15 were positive and one included zero.

Environmental associations

Weighting also changes other coefficients in the original four pairs. Landscape heterogeneity weakened, forest cover strengthened, and social vulnerability moved toward zero. Figure 8 reports all 12 slopes.

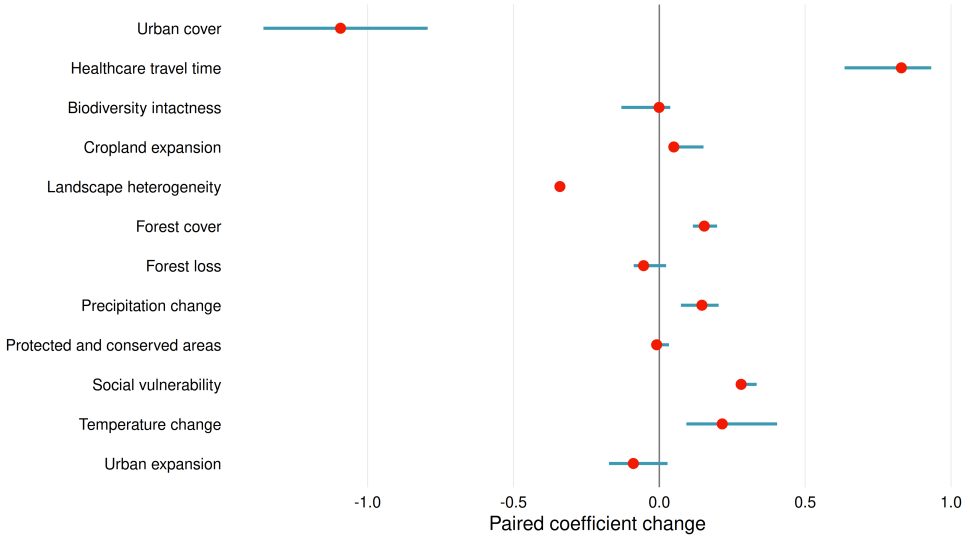


Figure 8. Original four-pair changes in posterior means under population-proportional weighting relative to the implemented log recipe. Red points show median within-pair changes; blue horizontal segments show the minimum and maximum across four draws. These ranges are not credible or confidence intervals. All coefficients use the same archived scaling in both arms.

Forest-cover medians increased from 0.340 to 0.494, with positive intervals in all fits. Forest-loss medians were negative under both recipes, changing from −0.072 to −0.126; intervals included zero in three log-recipe fits and two population-weighted fits. Forest cover measures state, whereas forest loss measures change.

With a planted environmental log-rate effect of 0.4, median fitted environmental coefficients were 0.396 versus 0.389 under rare occurrence and 0.358 versus 0.346 under saturation. Median paired changes were −0.003 and −0.004. The positive control therefore showed little weighting sensitivity in this coefficient. Under saturation, the generating log-rate effect and fitted occurrence coefficient have different interpretations.

West Nile incidence and dengue model sensitivity

The West Nile controls compared occurrence and population-offset incidence models using the same simulated cases across 3,084 continental US counties in 2004–2020. Both generating designs imposed constant individual risk and reporting. Occurrence models produced positive urban coefficients, whereas incidence estimates were centered near zero (Table 6).

Table 6. Urban posterior-mean medians; interval counts list occurrence first and incidence second. The outcome and coefficient scales differ between models.

Generating design	Occurrence median	Incidence median	Intervals including zero
Poisson null n = 50	0.297	0.002	0/50; 50/50
Joint negative binomial n = 20	0.068	0.000	3/20; 20/20

The observed West Nile incidence model retained a positive urban coefficient of 0.111 (95% credible interval 0.045–0.176). On a fixed county frame, changing logit to complementary log log retained a positive coefficient (0.766 to 0.559); adding a population offset at the same link gave approximately zero. In dengue, the urban and travel-time coefficient directions persisted across the prior and mesh alternatives (Figure 9).

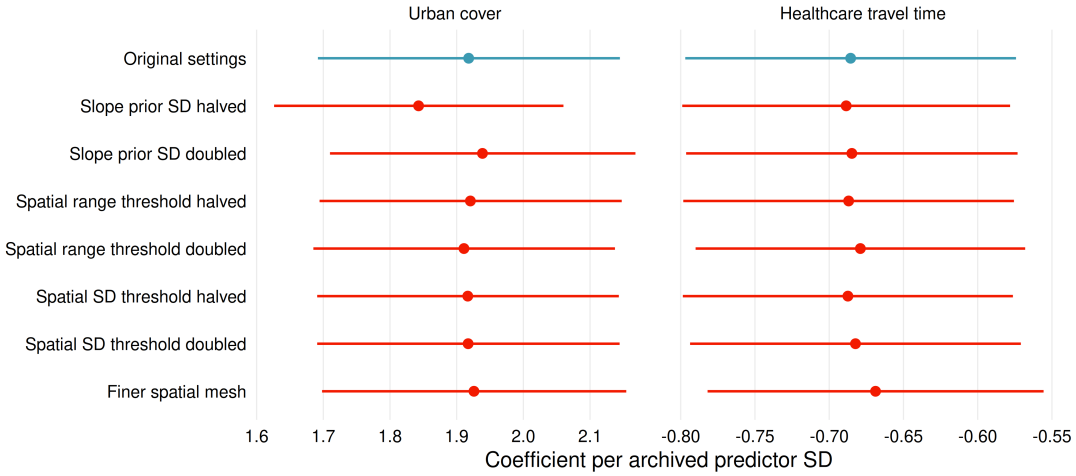


Figure 9. Archived-input dengue prior and mesh checks. Points and segments show posterior means and 95% credible intervals for eight settings: original (blue) and alternatives (red). The intercept prior remains fixed in slope-prior tests. All directions persist. These tests address different assumptions from background weighting.

Implications

Gibb et al. use population-weighted backgrounds to account for population exposure [1]. The simulations show that logarithmic weighting can leave associations with urban cover and healthcare access even when individual risk and reporting are constant. The controlled dengue comparison shows that this choice materially changes the fitted coefficients. These occurrence-versus-background associations do not separately identify exposure, infection processes and reporting, or establish their relative causal importance.

Population-proportional weighting is a useful comparator, but saturation shows that it is not a universal null for an outcome defined as at least one case. The empirical evidence concerns reconstructed dengue data: four pairs isolate logarithmic weighting, and 16 additional pairs assess the complete recipe. Positive urban associations persist, and the environmental positive control shows

little weighting sensitivity. These findings do not establish bias across diseases, test the cross-disease healthcare estimate or quantify the effect of temporal mismatch.

Methods

Synthetic landscape and occurrence generation

We used one fictional 60 × 60 grid of 1 km² cells, generated with seed 28092026. Let standard(x) subtract the mean and divide by the sample standard deviation across all 3,600 cells. With independent smooth spatial fields F_1, …, F_4 and independent standard-normal cell noise ε_1, …, ε_4, the covariates and population are:

$U = \text{standard}[\text{logistic}(0.7F_1 + 0.65\varepsilon_1)]$

$H = \exp(0.3 - 0.45U + 0.6F_2 + 0.5\varepsilon_2)$

$T = \text{standard}[\log(1 + H)]$

$E = \text{standard}(0.3U + 0.65F_3 + 0.45\varepsilon_3)$

$P = \max\{1, \text{round}[\exp(4.5 + 0.7U - 0.7T + 0.3F_4 + 0.15\varepsilon_4)]\}$.

U is standardized urban cover; H is travel time in hours; T is its standardized log transform; E is a standardized environmental covariate; and P is population per cell. The logistic function is 1/[1 + exp(−x)]. Each F is a standardized sum of eight Gaussian spatial bumps, with centers uniform over the landscape, standard-normal amplitudes and widths uniform from 4 to 16 km.

This construction makes population greater in more urban, more accessible cells. Realized populations range from 2 to 1,111. All cells are inhabited and equal in area, excluding area and empty-cell treatment as explanations for differences between synthetic sampling arms.

For cell i, exposure is $v_i = P_i \exp(\beta E_i)$, where the generating environmental log-rate effect β is 0 for the null and 0.4 for the positive control. Reporting probability q is 0.4. The baseline reported rate is $r = -\log(1 - m)/\max(v)$, with maximum occurrence probability m = 0.02 for rare occurrence or 0.80 for saturation. The baseline individual infection rate is $\lambda_0 = r/q$.

Expected reported counts are $\mu_i = P_i\lambda_0 q \exp(\beta E_i)$, giving occurrence probability $p_i = 1 - \exp(-\mu_i)$. Individual risk and reporting are spatially constant when β = 0. In the positive control, only E changes individual risk.

Each dataset conditions on 1,500 occurrence records, with cell counts drawn from a multinomial distribution with probabilities $p_i/\Sigma p_i$. Repeated selections represent independent observation windows conditional on occurrence. Fixing the record total keeps sample size constant between the rare and saturating scenarios.

We drew 3,000 backgrounds with replacement using normalized weights $\log(1 + P_i)$ or P_i. The null scenarios also used the known generating p_i as a diagnostic comparator. Shared uniform random numbers applied to each cumulative distribution paired the sampling arms. All arms within a replicate used identical occurrence records, covariates, scaling, mesh and priors.

Fifty replicates in each of four scenarios, rare/null, saturating/null, rare/positive and saturating/positive, produced 200 datasets and 500 fits. With scenario index $s = 1, \ldots, 4$ in that order and replicate $j = 1, \ldots, 50$, case seeds were $910000 + 10000s + j$ and background seeds were $920000 + 10000s + j$. Scenarios were summarized separately on the single fixed landscape.

Synthetic occurrence model

Let k_i and b_i denote occurrence and background record counts in cell i. The fitted likelihood is $k_i \sim \text{Binomial}(k_i + b_i, \pi_i)$, with

$$\text{logit}(\pi_i) = \alpha + \beta_u U_i + \beta_t T_i + \beta_e E_i + u(s_i).$$

Here π_i is the probability that a sampled record is labeled occurrence rather than background; α is the intercept; β_u, β_t and β_e are fitted slopes; and $u(s_i)$ is the spatial effect at coordinates s_i. The fitted environmental slope is distinct from the generating log-rate effect β. Grouping repeated records by cell preserves the binomial likelihood; cells receiving no records do not contribute.

Fixed coefficients and the intercept have independent normal priors with mean 0 and variance 1. The Matérn spatial effect uses a 295-vertex mesh, with inner/outer maximum edges of 8/20 km, a 4 km cutoff and inner/outer extensions of 8/20 km. Penalized-complexity priors assign probability 0.1 to spatial range below 30 km and probability 0.01 to spatial standard deviation above 3. Hyperparameters are estimated in each fit.

If occurrence records are sampled in proportion to p_i and backgrounds in proportion to w_i, sampled-label odds satisfy $\text{logit}\ \Pr(S = 1 \mid i) = \log K + \log p_i - \log w_i$. S denotes the record label, and K contains location-invariant sampling and normalization terms. Under rare occurrence and constant risk/reporting, $p_i \approx P_i \lambda_0 q$, so population weights approximately cancel the exposure gradient. Logarithmic weights leave a residual gradient, and saturation breaks proportionality to population. Nonzero slopes under the null can therefore describe the sampled-label contrast without identifying a reporting mechanism.

Dengue data and background sampling

We used the archived majority-rule dengue specification [2], with 11,313 global occurrence records, 12,108 backgrounds per fit and 12 predictors. The sampling frame is the masked WorldPop 2010

population-count raster. Cells with finite, nonnegative population are eligible; cells with missing population are excluded. Cases, covariate surfaces and model components remain fixed across arms.

Let P_i be persons in cell i and A_i its area from terra::cellSize. Arm A assigns weight $A_i \log(1 + P_i)$ to inhabited cells and $A_i c$ to empty cells, with c = 1.06840784199454, the median positive log weight. B assigns empty cells zero weight while keeping the area and log factors. C uses $A_i P_i$, and D uses P_i alone. Each weight vector is normalized over its eligible cells.

B minus A tests the empty-cell rule under area-weighted log sampling; C minus B isolates log versus linear weighting; D minus C removes the area factor under linear weighting. D minus A compares complete recipes. These sequential contrasts are conditional on the other choices, rather than factorial main effects. Because P_i is a population count, C retains an additional area factor and D is population proportional.

Shared uniforms select cells through inverse cumulative probabilities in a fixed raster order. Additional shared uniforms jitter longitude and latitude within each selected cell. Sampling is with replacement, and jittered points are not clipped again to the study polygon. The original four seeds are 980019, 980027, 980035 and 980043.

Background covariates are extracted over geodesic buffers of radius 5,035.926 m. Original case polygons and coordinates remain fixed. Exact duplicate case geometries are extracted once and reused: 9,478 unique polygons supply 11,313 separate outcome records.

All four arms were fitted for each original seed. Sixteen additional A/D pairs used seeds 980001–980016, fixed before fitting, with the same cases, surfaces, extraction rules and model settings. The controlled B/C comparison remains based on four pairs.

Covariate reconstruction

All 12 covariates were reconstructed consistently for cases and new backgrounds. Archived case values were not mixed with reconstructed background values in the weighting comparisons.

Covariates are polygon means weighted by raster-cell coverage fractions using exactextractr, without population or additional geodesic area weights. Polygons are transformed to each raster's coordinate system; covariate grids retain their own resolutions. Dateline-crossing polygons are wrapped before extraction.

Predictor group	Source and processing used
Forest and urban cover	Copernicus 2015, collection 3, v3.0.1 tree-cover and built-up fractions. The no-data code 255 is excluded; extracted percentages are divided by 100.
Cropland expansion	GLAD Global_cropland_3km_netgain product; percentages become fractions using the same mean-then-divide convention.
Biodiversity intactness	Newbold et al. (2016), representing 2005. The reconstruction restores Float64 values and an explicit EPSG:4326 reference.

Predictor group	Source and processing used
Landscape heterogeneity	EarthEnv EVI dissimilarity. Values above 200000 become missing; valid values are multiplied by 0.0001 and passed through Float32 before extraction.
Healthcare travel time	MAP motorized 2020 travel time in minutes. Extract the mean first, then calculate log(1 + mean minutes).
Urban expansion	ESA 2000 v2.0.7cds and 2018 C3S v2.1.1 land cover. Native-grid indicator of nonurban-to-urban class 190; baseline water class 210 is missing.

Urban expansion retains its native 1/360° grid. Five predictors have unresolved source or processing differences: temperature change, precipitation change, protected areas, forest loss and social vulnerability.

Temperature and precipitation use ERA5-Land monthly means for 1950–1970 and 2001–2020. Temperature is converted to degrees Celsius and averaged with equal weight across each year's 12 months. Precipitation is converted from daily mean meters to millimeters, multiplied by days per month and summed annually. Change is the recent-period mean minus the baseline mean.

Annual rasters use Float32 values on the native 0.1° grid, with no interior resampling. Cells with baseline precipitation above 10,000 mm/year are masked. A periodic longitude column permits extraction at the antimeridian. These reconstructed climate values do not match the authors' processed values.

Protected-area coverage uses October 2026 WDPCA protected and conserved polygons, including other effective area-based conservation measures, in place of the September 2022 snapshot. Polygons are rasterized as a binary union on the retained template grid using the pixel-center rule, without geometry simplification.

Forest loss uses Hansen 2021 v1.9: tree-cover percentage in 2000 multiplied by an indicator of recorded loss during 2001–2021, divided by 100. The native product is averaged to 0.01° cells before polygon extraction. Equivalence to the authors' Earth Engine processing is unverified.

Social vulnerability uses a GRDI layer aggregated by successive means with factors 10 and 2, without intermediate rounding. Numerical agreement is close, but the historical source edition remains unconfirmed.

All arms use $z = (x - a)/d$, where a and d are the mean and standard deviation from the archived combined case/background frame. Scaling is not recomputed after resampling. Healthcare travel time is log(1 + mean minutes) before standardization. Table M1 compares archived and reconstructed case values where both are finite.

Table M1. Root mean squared error (RMSE) between saved reconstructed and archived case values, divided by the archived predictor standard deviation (SD). Missingness disagreements count records finite in one matrix and missing in the other, among 11,313 cases. Small error indicates numerical agreement, not confirmed source provenance. Climate, forest-loss and protected-area discrepancies remain material.

Case predictor	RMSE / archived SD	Missingness disagreements
Biodiversity intactness	0.022	0
Cropland expansion	0.001	0
Landscape heterogeneity	0.008	0
Forest cover	0.016	0
Forest loss	0.346	146
Precipitation change	0.465	2
Protected areas	0.324	0
Social vulnerability	<0.001	0
Temperature change	0.261	0
Urban cover	<0.001	0
Urban expansion	<0.001	0
Healthcare travel time	<0.001	0

Dengue model

Occurrence records receive y = 1 and backgrounds y = 0. The Bernoulli model is logit $\Pr(y = 1) = \alpha + \Sigma\beta_k z_k + u(s)$, with intercept α, standardized covariates z_k, slopes β_k and spatial effect u(s) at location s. There is no population offset. The outcome distinguishes sampled record types rather than estimating individual infection probability.

The archived Matérn mesh has 5,026 vertices and geographic-degree coordinates. Spatial penalized-complexity priors assign probability 0.1 to range below 100 coordinate units and probability 0.01 to spatial standard deviation above 10. The range threshold is not 100 km. Fixed coefficients, including the named intercept, have independent normal priors with mean 0 and variance 1.

All fits use R 4.3.3 and INLA 23.03.26, experimental mode, adaptive strategy, cmin = 0 and one thread. Hyperparameters are re-estimated in each fit.

Missing covariates pass through the archived INLA pipeline without deleting incomplete records or adding an imputation model. Among reconstructed cases, missing values range from zero to 344 per predictor. Thirteen case rows have zero spatial-projector support on the archived mesh and remain in the likelihood without a spatial contribution; backgrounds outside mesh support are treated the same way.

Archived value restoration and predictor omission

At the original case and background locations, we fitted the full model using archived and reconstructed covariates. Five further specifications restored archived values for one unresolved

predictor at a time, and a sixth restored all five. Restorations apply to both cases and backgrounds, retaining all other reconstructed values, archived scaling, mesh and model settings.

We also omitted the five unresolved predictors together and fitted all four sampling arms at all four original seeds. The remaining seven predictors, observations, sampled locations and model settings were held fixed. This tests sensitivity to the adjustment set; archived-value restoration assesses reconstruction at the original locations.

Prior and mesh sensitivity

Prior and mesh comparisons use archived dengue covariates and locations. Eight specifications comprise the original model, slope-prior standard deviations of 0.5 or 2, spatial range thresholds of 50 or 200, spatial standard-deviation thresholds of 5 or 20, and a finer mesh. The intercept prior remains normal with mean 0 and variance 1; spatial prior probabilities remain 0.1 for range and 0.01 for standard deviation.

The finer mesh multiplies the original maximum edges, cutoff and inner extension by 0.75, holding the outer extension fixed. This increases the number of vertices from 5,026 to 7,703.

Country omissions

Cases and backgrounds are assigned to Brazil, India or the Philippines by intersecting their modeled coordinates with GADM 4.1 ADM0 boundaries using sf with s2 enabled. Boundary points are included. One invalid India feature is repaired with st_make_valid, and all six features labeled India in the source are retained. Coordinates outside boundaries remain included, without snapping or administrative-label overrides.

Original country labels are missing. Exclusions therefore refer to modeled geography, removing 2,773 case coordinates for Brazil, 899 for India and 698 for the Philippines, leaving 8,540, 10,414 and 10,615 cases, respectively.

Each exclusion filters the existing global backgrounds without redrawing or topping up samples or re-extracting covariates. Background counts vary by arm and seed, and a retained buffer may cross the excluded boundary. Covariates, archived scaling, model, priors and global mesh remain unchanged. All 48 fits, three omissions × four seeds × four arms, are included. These comparisons share draws and overlapping records; the global comparison retains equal background counts across arms.

West Nile simulations and models

West Nile controls use archived geography, static population and covariates for 3,084 continental US counties over 17 years, 2004–2020. The seven predictors are biodiversity intactness, forest loss, landscape heterogeneity, log livestock density, precipitation change, temperature change and urban cover.

The Poisson design gives county i an annual reported count with mean $\mu_i = P_i r$, where P_i is population and $r = 2.492283 \times 10^{-6}$ is the common reported rate per person-year. We chose r so that $\Sigma 17[1 - \exp(-P_i r)] = 7{,}446$, matching positive county-years in expectation. The expected case total is approximately 12,516, compared with 35,233 observed cases. Fifty replicates use seeds 260927 + j, j = 1, …, 50.

The joint design uses negative-binomial annual counts with mean $\mu_i = P_i r^*$ and size κ. The rate $r^* = 7.016047 \times 10^{-6}$ matches 35,233 expected cases, and $\kappa = 0.1544533$ matches 7,446 expected positive county-years through $\Sigma 17[1 - (1 + P_i r^*/\kappa)^{-\kappa}]$. Variance is $\mu_i + \mu_i^2/\kappa$. Twenty replicates use seeds 660927 + j. Totals are matched in expectation, not fixed in each replicate.

For each county, we first draw the number of positive years from Binomial(17, $1 - p_{0i}$), where p_{0i} is the generating count distribution's zero probability. Positive annual counts are then sampled conditional on exceeding zero and summed. Each replicate supplies the same positive-year counts and cumulative cases to the occurrence and incidence fits. Both designs impose constant reported risk per person; reporting probability is absorbed into the rate.

Occurrence fits use positive county-years as repeated occurrence records alongside 9,000 fixed archived backgrounds. Cases are grouped with binomial totals equal to their positive-year counts; each background contributes a zero label and one trial. The archived spatial logit model uses a 3,109-vertex mesh, normal coefficient priors with variance 1, and spatial prior thresholds of 20 geographic coordinate units for range and 5 for standard deviation, at probabilities 0.1 and 0.01.

Predictors are centered and scaled within the combined sample using positive-year multiplicities for cases and unit weights for backgrounds. Missing standardized entries are replaced by zero. Slopes are then rescaled to archived predictor units. The background frame is fixed across replicates.

Incidence fits retain the authors' negative-binomial model and seven predictors. The response is cumulative county cases, with offset $\log(P_i/100000)$; the common 17-year period is absorbed by the intercept. County ordering, scaling and spatial components are retained. Negative-binomial size is estimated with the archived pc.mgamma prior, parameter 7. Archived hyperparameters initialize optimization with restart enabled.

Separate link/offset controls use the complete county frame, including zero years, and model positive years out of 17. With data, scaling, mesh and priors fixed, we compare logit without an offset, complementary log log without an offset, and complementary log log with offset $\log(P_i/100000)$. Under Poisson annual counts, $\log[-\log(1 - p_i)] = \log P_i + \log r$. The offset therefore accounts for population exposure at a fixed link and observation frame.

Statistical summaries

Individual-model summaries use posterior means and the 2.5th and 97.5th marginal posterior percentiles. Simulations summarize the distribution of means and count intervals containing zero. For an occurrence-versus-background slope, this count is not automatically coverage of an individual-risk parameter.

Paired changes are differences between posterior means under the specified sampling arms, calculated within each seed before summarizing. We report means, medians, sample standard deviations and observed ranges. Monte Carlo standard errors for mean changes are $s/\sqrt{n}$, where s is the sample standard deviation and n is the number of pairs. Empirical quantiles use R type 7 interpolation.

Across-draw variation describes background sampling conditional on the fixed cases, reconstructed surfaces and model. It is distinct from each fit's marginal credible interval and does not include reconstruction uncertainty. A posterior interval for a paired change cannot be obtained by subtracting marginal interval endpoints. All 12 dengue slopes are reported.

Prepared covariate matrices, spatial meshes, sampling seeds, model code and saved coefficient estimates are supplied in the replication package. Its README gives commands to regenerate figures and tables or refit the models. Reproduction starts from those prepared inputs; raw GIS sources are not bundled.

References

[1] Gibb et al. (2026). The anthropogenic fingerprint on emerging infectious diseases. Nature. doi:10.1038/s41586-026-11058-6.

[2] Archived analysis code and outputs, version 1.0.0. doi:10.5281/zenodo.20738794.